

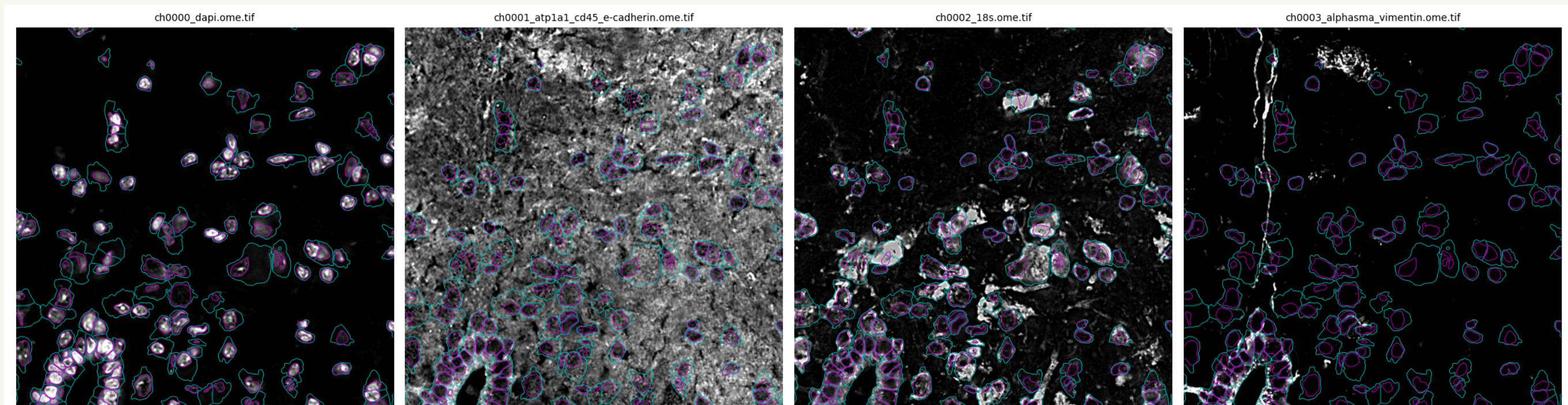
■ BEYOND BOUNDARIES · LAB MEETING · 7 OCT 2026

What does the Xenium *multimodal segmentation kit* tell us that the transcriptome doesn't?

Mouse EAE spinal cord, RRMAP2 runs 5 + 6 · 18 Xenium 5K images · 51 tissue pieces · 25 animals · 500,379 annotated cells · 201 image features per cell

THE QUESTION

Four stains are imaged anyway, to draw the cell masks. *Is there biology in them?*



DAPI · ATP1A1/CD45/E-Cad (boundary) · 18S rRNA (interior; drew ~95 % of masks) · αSMA/Vimentin. Per cell: intensity in nucleus / cytoplasm / rim / surrounding ring, texture, polarity, shape. **Vim, Acta2 and Atp1a1 are not on the 5K panel.**

■ BOTTOM LINE

Three kinds of added information. *No hidden cell states.*

Protein

...for genes not on the panel

Vimentin, α SMA, ATP1A1 per cell. Astrocyte vimentin lags reactive RNA in active disease.

Tissue

...context around each cell

White-matter neuropil loss in lesions ($\sim 25\%$).
Anatomy that adds to the transcriptome (0.68 $\rightarrow$ 0.73).

Images alone

...recognise cells and map lesions

Cell type 46 % (chance 7 %), lesion maps AUROC 0.90, and this holds **across imaging runs**.

Not added: CD45 (undetectable in mouse cord); lesion identity and clinical score beyond what the transcriptome already gives. $\sim 85\%$ of image variance is unexplained by RNA, but that's mostly optics.

DO THE STAINS MEASURE WHAT THEY CLAIM?

Known biology: yes, in every animal. *CD45: no.*

0.86

ependymal vimentin

0.78

neuronal 18S (cytoplasm)

0.73

leukocyte vimentin >
neuron / oligo

0.66

VSMC αSMA-bright

0.48

leukocyte CD45 at the rim:
chance; boundary channel
= ATP1A1 neuropil only

AUROC; positives > 0.5 in 25/25 animals; also in boundary-segmented cells. → Ask 10x whether the kit's CD45 antibody is mouse-reactive.

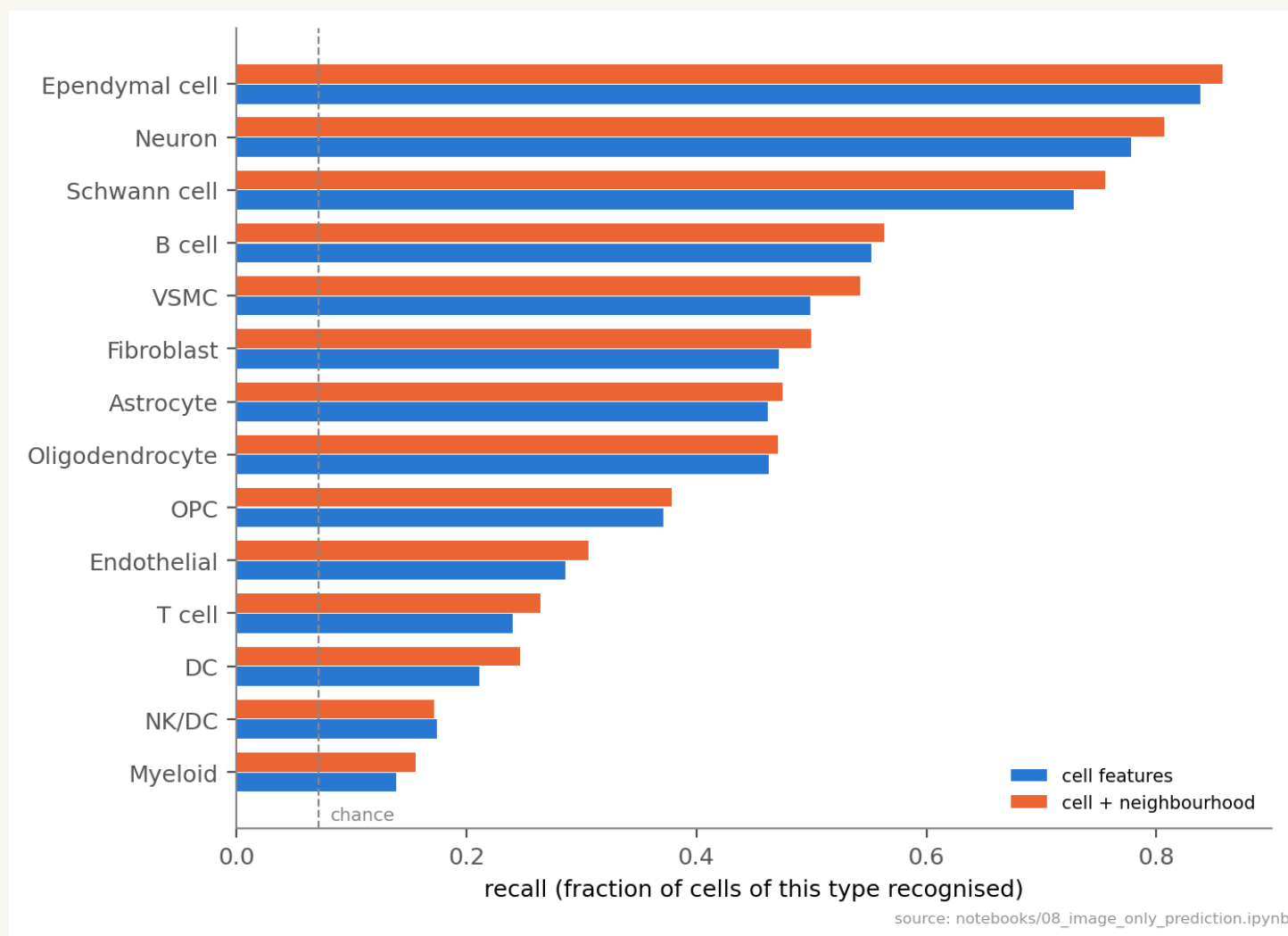
■ HOW MUCH OF THE IMAGE IS NEW?

Most of the image isn't in the transcriptome, *but most of that is optics*

- Covariates + 50 transcriptome PCs explain a median **~15 %** of each image feature (best: nucleus:cell ratio, rim/cyto intensity).
- The remainder is spatially coherent, but largely a **local field shared across cell types** ($r \approx 0.38$; follows neuropil glow, not lesions).
- The cell-specific remainder barely moves with lesion state: 19/924 tests, ~ 0.1 SD.
- Residual clustering: no discrete hidden states, only continuous gradients.

The kit is not a source of new *cell types*. It's a source of new *measurements* for specific questions.

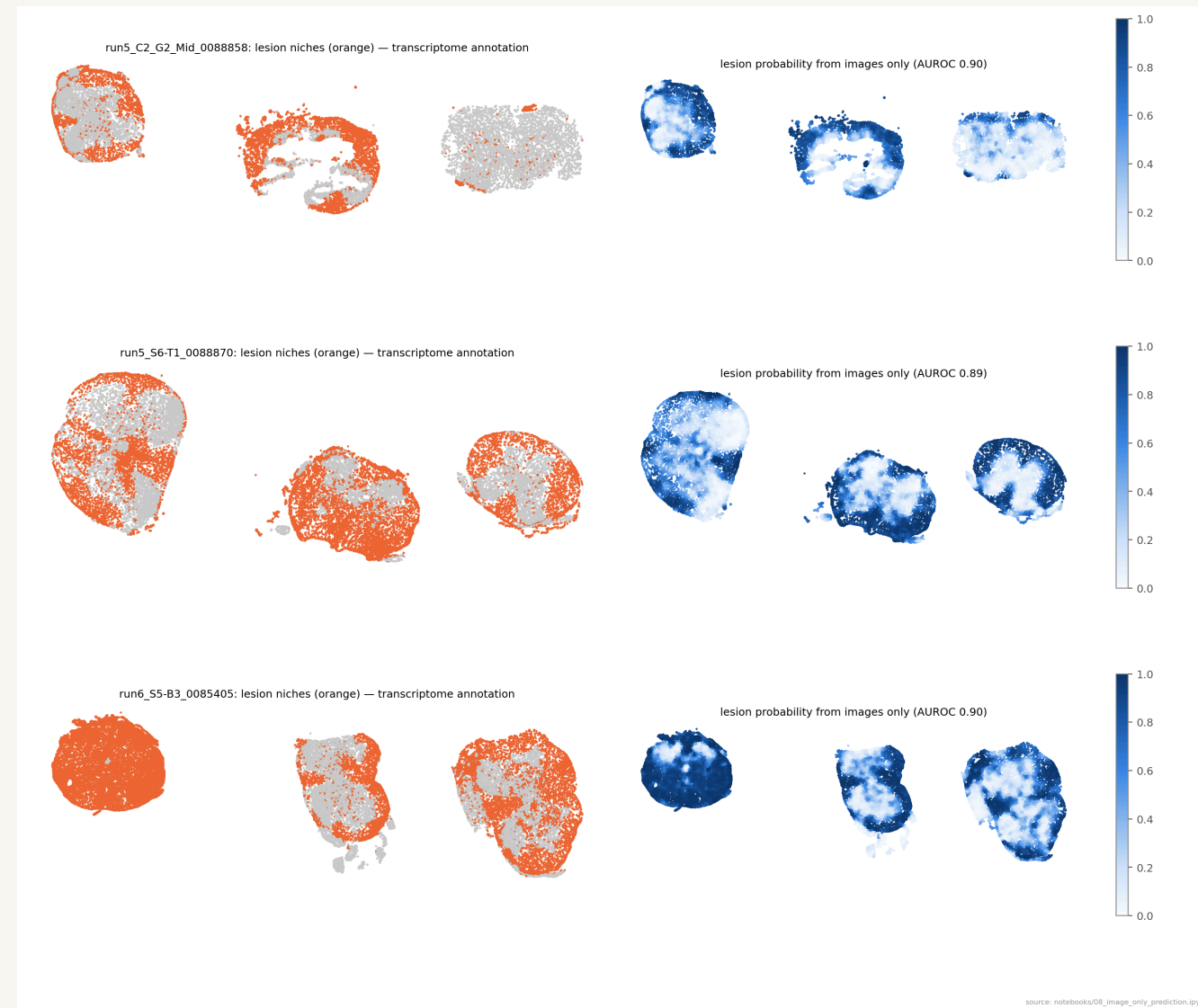
IMAGES ALONE, HELD-OUT ANIMALS



No transcripts: **46 %** of 14 cell types
(chance 7 %)

- Ependymal 0.86 · neuron 0.81 · Schwann 0.76
- Immune poor (myeloid 0.16, T 0.26): no CD45
- Errors stay within lineage
- Subtypes: microglia 72 %, reactive astrocytes 68 %, cholinergic neurons 72 %
- Identity is spread across all four stains, so it's not an artefact of the 18S masks

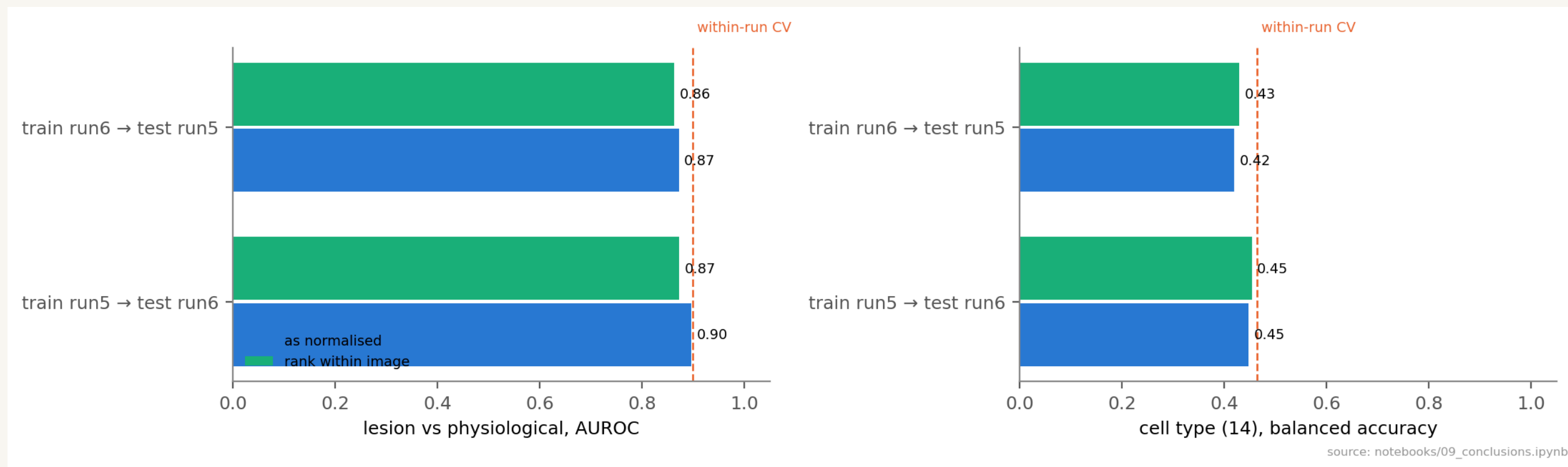
Lesion maps from images: *AUROC 0.90* (transcriptome neighbourhood 0.92)



Top signal: vimentin-channel texture/intensity in surrounding tissue. Anatomy (10 regions): images 0.63, transcriptome 0.68, both 0.73.

NEW TODAY · DOES IT SURVIVE A NEW RUN?

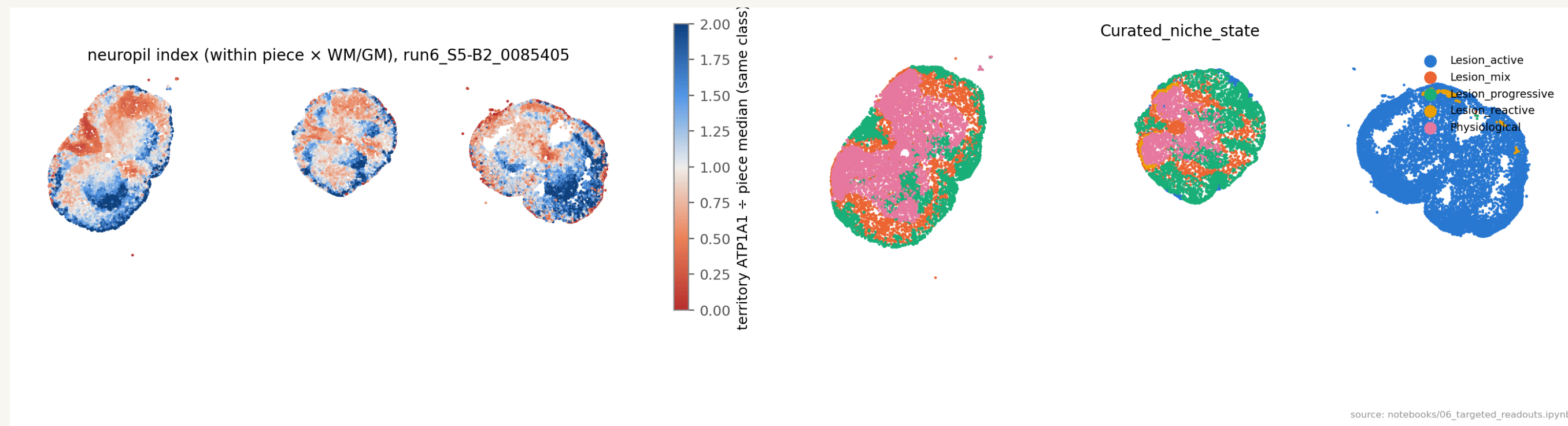
Images identify their run with 100 % accuracy, *yet the models still transfer*



- Train run5 → test run6: lesion AUROC **0.90** (per animal 0.91); run6 → run5 **0.87**. Within-run: 0.90.
- Cell type 0.42–0.45 vs 0.46 within-run. Per-image rank transform doesn't help, so per-image normalisation is enough.

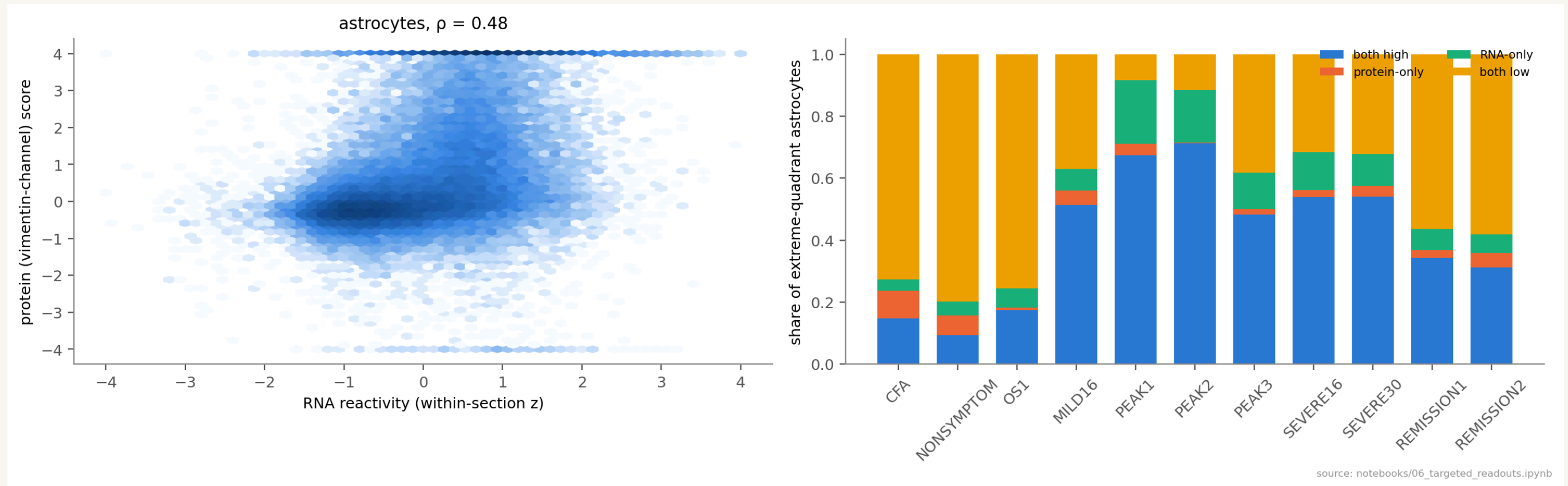
BIOLOGY ONLY THE IMAGES SHOW · 1

White-matter lesions lose **25 %** of the *ATP1A1* around each cell



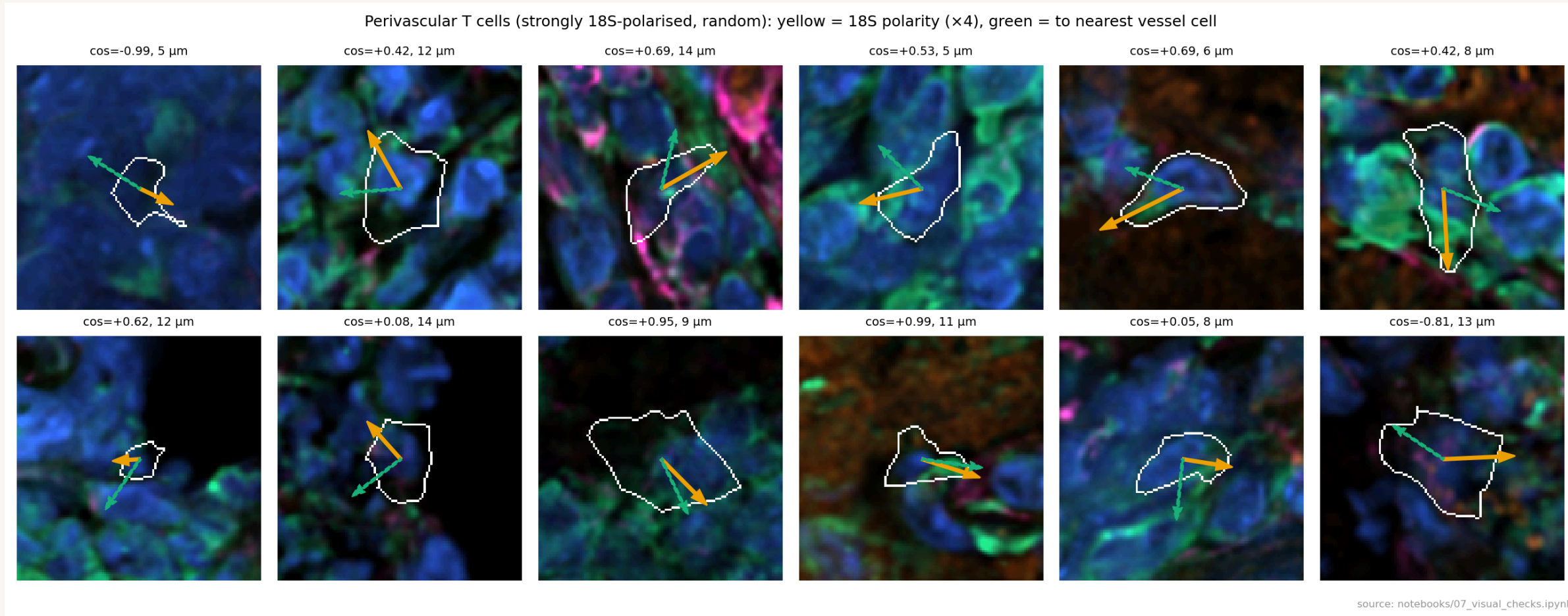
Lesion vs healthy WM **within the same piece**: 38 pieces, $p = 8 \times 10^{-6}$. Grey matter: no change ($p = 0.62$). This is tissue damage the cell's own transcriptome can't carry. Not related to clinical score.

● Astrocyte reactive RNA *precedes* vimentin protein



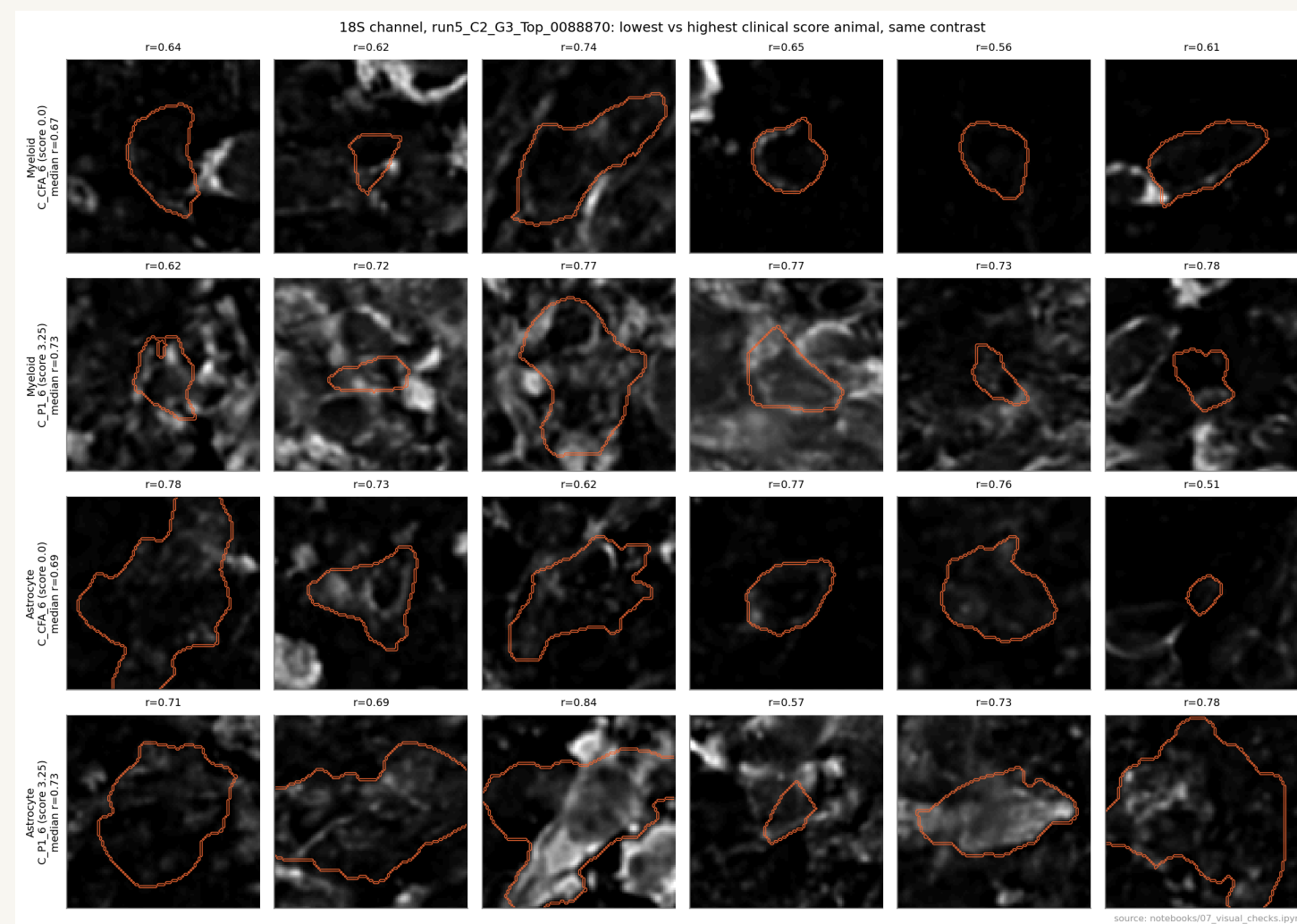
- Protein vs RNA reactivity $\rho = 0.48$ (C3 0.50, Gfap 0.38; Vim itself isn't on the panel)
- **RNA-only** astrocytes peak in active disease (PEAK1 20 %, CFA 3.5 %), fall in remission (active vs remission $p = 0.019$)

18S rRNA: activation, and *T cells facing the vessel*



- 18S polarity points vessel-wards in T cells (cos 0.13; 0.20 in the most polarised quarter), fibroblasts and MDM, but not in neurons or oligodendrocytes, so it's not blur
- Lesion 18S rises beyond RNA content in activated cells (Schwann +0.8, endothelium +0.4 SD); the oligodendrocyte drop is just RNA loss

CANDIDATE · NEEDS REPLICATION



Smoother 18S texture in *sicker* animals

- Rises with clinical score in 6 cell types
- Holds **within section** (p 0.53–0.86) and after a crowding adjustment (0.41–0.81)
- DAPI texture (same optics) doesn't follow, so it's probably not focus
- Ribosome redistribution / RNA degradation, or tissue handling per animal

Animal level: score from images p 0.62 vs transcriptome 0.79. Images are informative but redundant.

TAKE-HOME

How to use the segmentation stains

Use them for

Vimentin / αSMA / ATP1A1 protein (targets missing from the panel)

Tissue context: neuropil loss, anatomy, lesion maps

Protein-vs-RNA timing (astrocyte reactivity)

Per-cell 18S as a ribosome / activation readout (adjusted for transcript density)

QC: segmentation method, spill-over-prone neighbourhoods

Don't use them for

CD45 / immune membrane (undetectable in mouse cord)

Replacing the transcriptome for lesion identity or severity

Cell polarity from DAPI or αSMA/Vim (optical bleed)

Pooling *raw* feature values across runs (models transfer; raw values don't)

18S distribution features as biology in 18S-segmented cells

CAVEATS & NEXT STEPS

Caveats

- Lesion niches are transcriptome-defined, so image-vs-RNA comparisons for lesions are tilted toward RNA
- α SMA and vimentin share a channel
- 18S drew ~95 % of the masks
- 25 animals, 2 CFA pieces; 2 runs from one batch

Everything: notebooks/00_summary · 09_conclusions · report/BeyondBoundaries_summary.pdf

Next

- **Replicate on runs 1–3** (P drive): 18S texture, WM neuropil loss, astrocyte RNA → protein lag, T-cell polarity, image-model transfer
- Separate vimentin from α SMA (VSMC-adjacent subset); check CD45 reactivity with 10x
- Feed the neuropil-loss index and per-cell 18S into RRMAP2 lesion/niche analyses as covariates